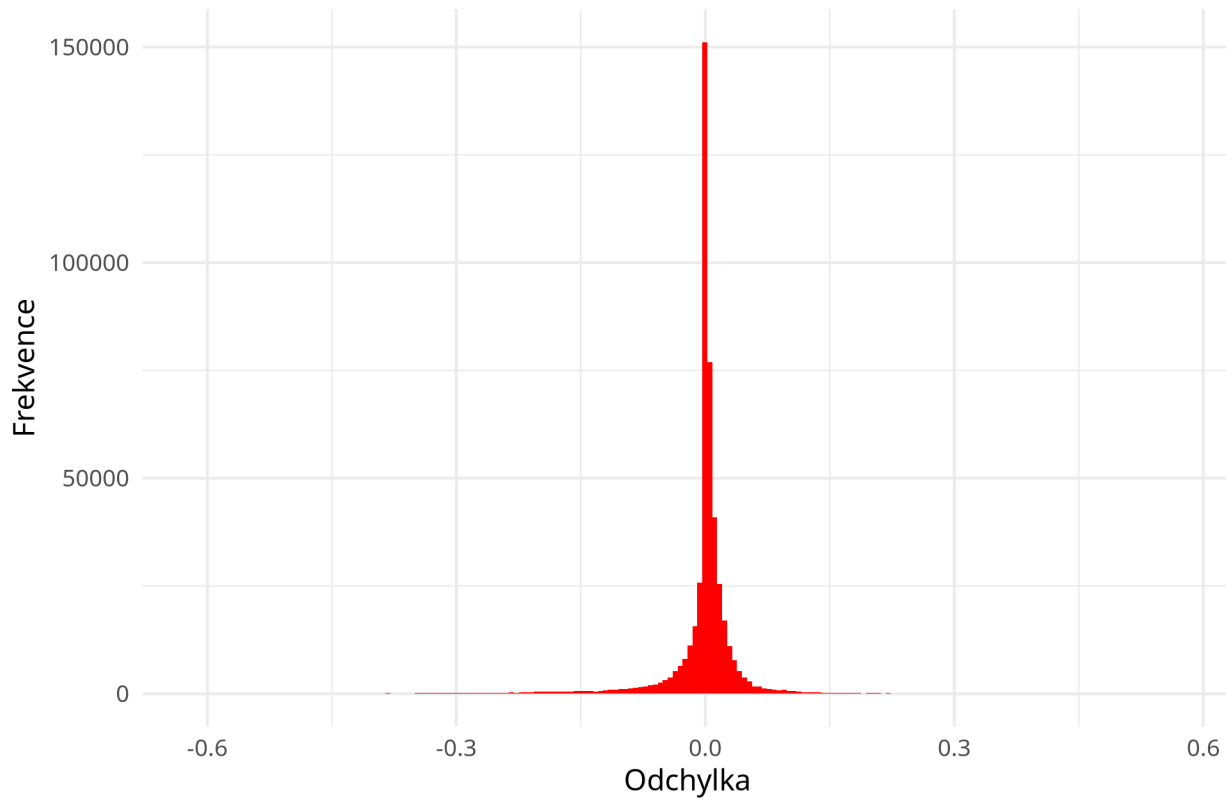


Regresní analýza

Základní data

Základními daty myslím data s opravami v předchozích iteracích analýzy, čili oprava dat u titulů NFLX a NVDA.

Histogram odchylek odhadované ceny od reality



Histogram počtu opcí s danými odchylkami od reality na ověření normálního rozdělení.

LMER model - základní

```
subset_args <- c(  
  "S",  
  "moneyness",  
  "K",  
  "r",  
  "sigma",  
  "q",  
  "T",  
  "am_eu",  
  "call_put",  
)
```

```

"model",
"ticker"
)
interactions <- c(
  "q:am_eu",
  "(1 | ticker)",
  "(1 | snapshot_month)"
)

rhs <- paste(c(subset_args, interactions), collapse = " + ")
formula <- as.formula(paste("pricing_error ~ ", rhs))

rmr_relative <- lmer(formula, data = train_data_rel)

## fixed-effect model matrix is rank deficient so dropping 2 columns / coefficients
summary(rmr_relative)

## Linear mixed model fit by REML ['lmerMod']
## Formula: pricing_error ~ S + moneyness + K + r + sigma + q + T + am_eu +
##      call_put + model + ticker + q:am_eu + (1 | ticker) + (1 |
##      snapshot_month)
##      Data: train_data_rel
##
## REML criterion at convergence: -1128760
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -11.7599  -0.1024   0.1051   0.2849  10.6498
##
## Random effects:
##      Groups             Name             Variance Std.Dev.
##      ticker              (Intercept)  1.385e-04  0.0117694
##      snapshot_month      (Intercept)  6.298e-07  0.0007936
##      Residual                                2.892e-03  0.0537730
## Number of obs: 375362, groups:  ticker, 18; snapshot_month, 4
##
## Fixed effects:
##              Estimate Std. Error t value
## (Intercept)  -7.265e-02  1.235e-02  -5.884
## S              1.479e-05  1.826e-06   8.099
## moneyness     -4.186e-06  1.085e-05  -0.386
## K             -2.254e-05  4.054e-07 -55.592
## r              2.813e+00  8.569e-02  32.832
## sigma        -5.614e-02  3.325e-03 -16.885
## q            -1.288e+00  1.548e-01  -8.319
## T              4.332e-03  1.264e-04  34.276
## am_eueuropean  2.667e-03  1.724e-02   0.155
## call_puts     -1.661e-02  1.785e-04 -93.033
## modeluser_crr -4.514e-05  2.151e-04  -0.210
## modeluser_mc   1.689e-04  2.150e-04   0.786
## ticker~VIX     -1.931e-02  1.791e-02  -1.078
## ticker~XDA     -4.662e-02  1.832e-02  -2.544
## ticker~XDB      8.185e-04  1.799e-02   0.046

```

```
## ticker^XDE      -2.002e-02  1.764e-02  -1.135
## ticker^XDN      -1.786e-02  1.783e-02  -1.002
## tickerAAPL      -6.936e-03  1.669e-02  -0.416
## tickerAMD        2.331e-03  1.665e-02   0.140
## tickerAMZN      -3.285e-03  1.667e-02  -0.197
## tickerGOOGL      7.275e-03  1.668e-02   0.436
## tickerJNJ        2.443e-02  1.750e-02   1.396
## tickerKO         2.489e-02  1.735e-02   1.434
## tickerMETA       2.757e-04  1.667e-02   0.017
## tickerMSFT      -7.633e-04  1.673e-02  -0.046
## tickerNFLX      -1.719e-02  1.667e-02  -1.031
## tickerNVDA       3.969e-03  1.666e-02   0.238
## tickerPG         2.428e-02  1.722e-02   1.410

##
## Correlation matrix not shown by default, as p = 28 > 12.
## Use print(x, correlation=TRUE) or
##      vcov(x)          if you need it

## fit warnings:
## fixed-effect model matrix is rank deficient so dropping 2 columns / coefficients
## Some predictor variables are on very different scales: consider rescaling.
## You may also use (g)lmerControl(autoscale = TRUE) to improve numerical stability.
r2(rmr_relative)

## Warning: Can't compute random effect variances. Some variance components equal
## zero. Your model may suffer from singularity (see `?lme4::isSingular`
## and `?performance::check_singularity`).
## Decrease the `tolerance` level to force the calculation of random effect
## variances, or impose priors on your random effects parameters (using
## packages like `brms` or `glmmTMB`).

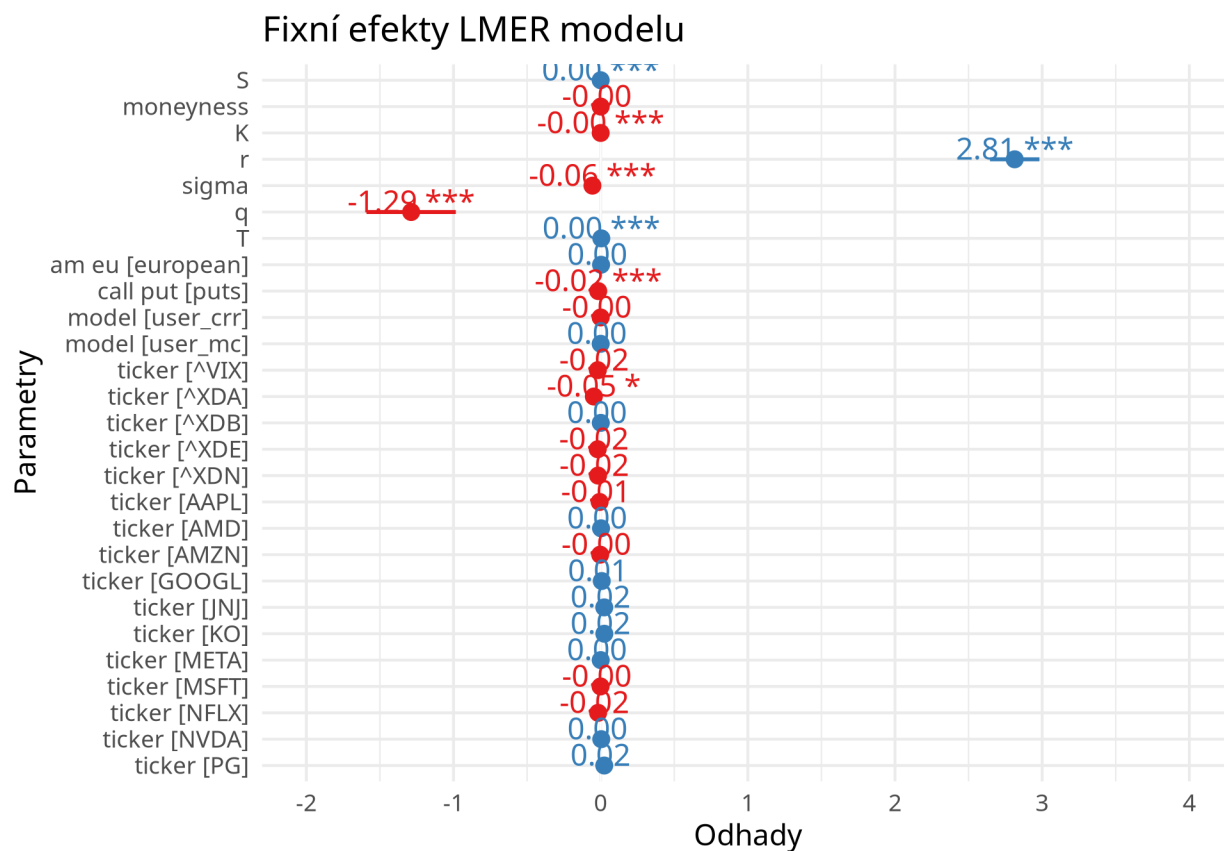
## Random effect variances not available. Returned R2 does not account for random effects.

## # R2 for Mixed Models
##
## Conditional R2: NA
## Marginal R2: 0.078
```

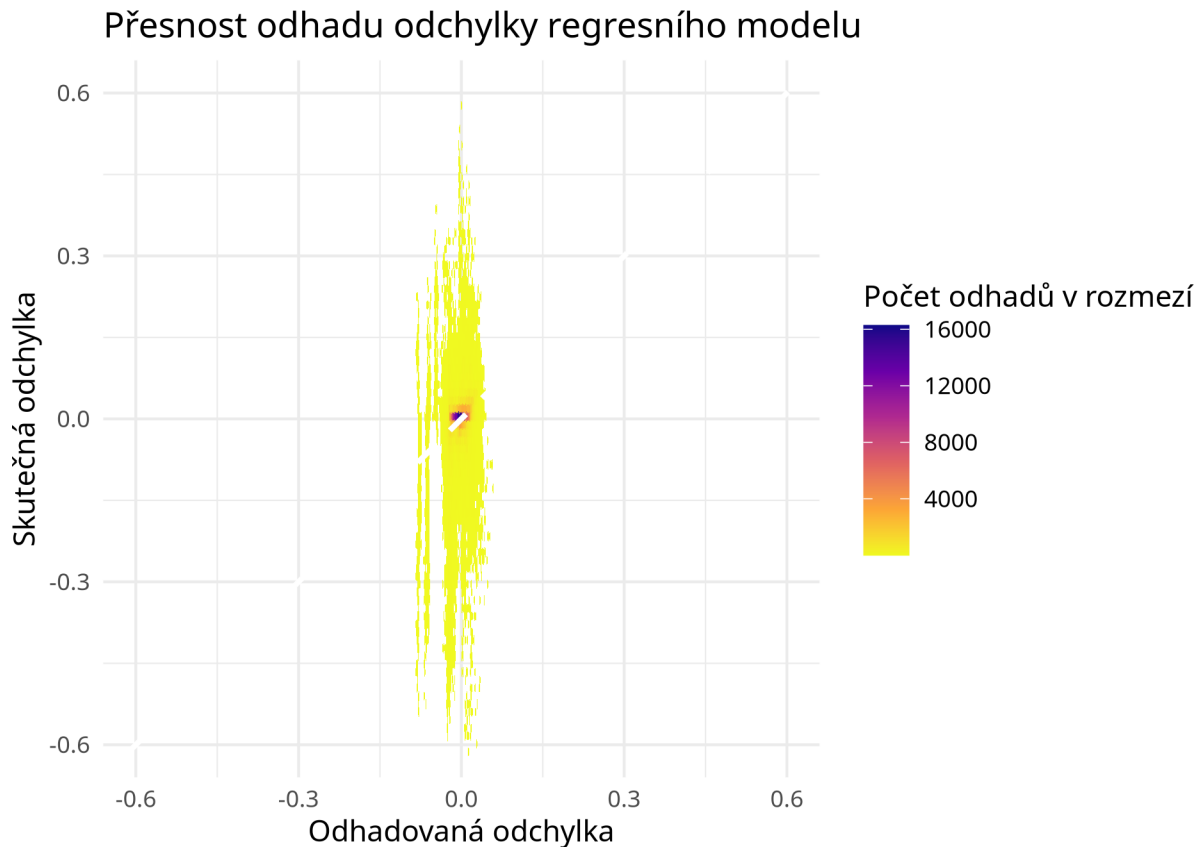
Model je singulární. Toto je způsobeno:

- přítomností parametru *moneyness* (S/K) a parametrů S a K ,
- přítomností parametru *ticker* ve fixních efektech a náhodných efektech.

Toto by neměl být problém pro model jako takový, ale znemožňuje to vypočítat podmíněné R^2 .



Poměr S/K , styl využití opce, použitý model a tituly jsou statisticky nevýznamné. Ostatní parametry jsou statisticky významné.



Samostatné clustery v levé části grafu jsou tvořeny opcemi na titul $\hat{VIX}$.

```
anova(rmr_relative)
```

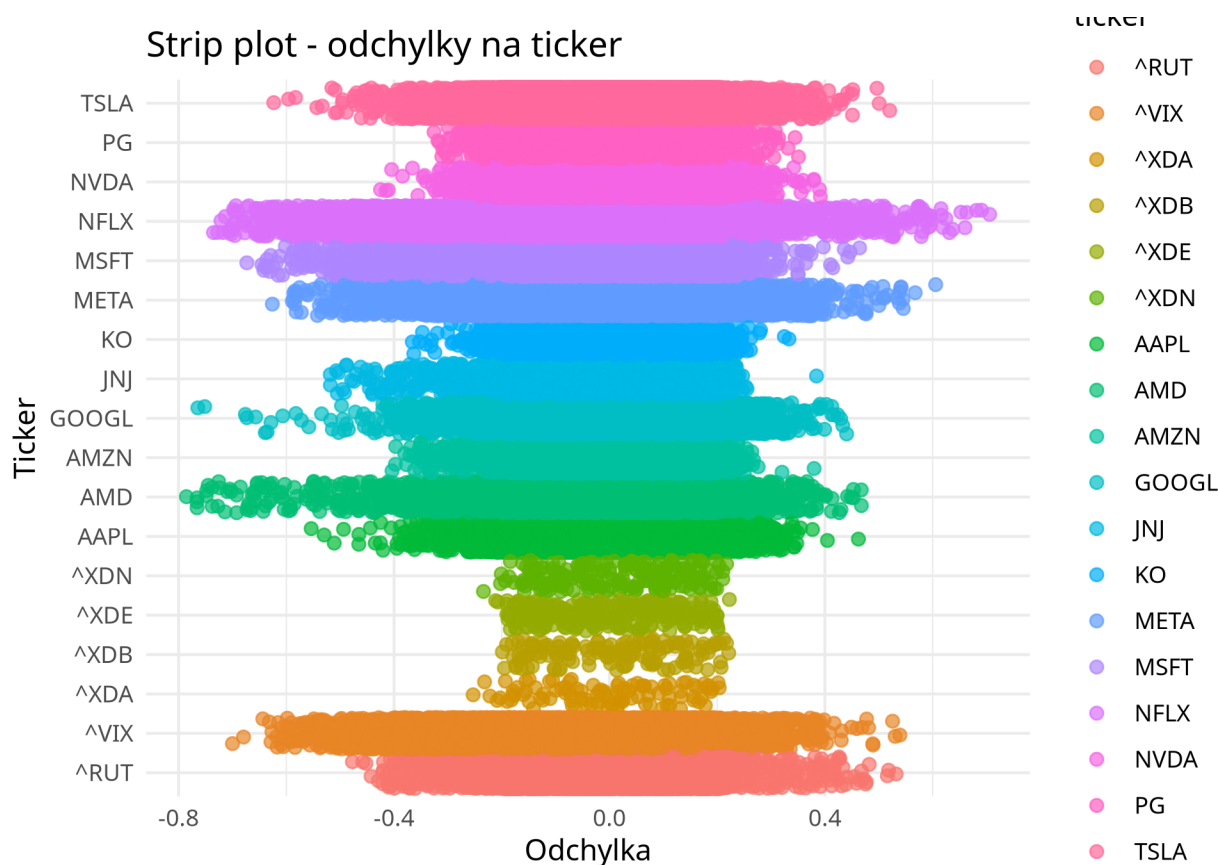
```
## Analysis of Variance Table
##          npar  Sum Sq Mean Sq  F value
## S              1  0.0088  0.0088    3.0507
## moneyness      1  0.3009  0.3009   104.0737
## K              1  4.8301  4.8301  1670.4253
## r              1  2.2843  2.2843   789.9817
## sigma         1  0.9104  0.9104   314.8502
## q             1  0.1754  0.1754    60.6659
## T             1  3.2291  3.2291  1116.7489
## am_eu         1  0.0218  0.0218     7.5559
## call_put      1 25.0151 25.0151  8651.1409
## model         2  0.0032  0.0016     0.5521
## ticker       16  0.0579  0.0036     1.2521
```

Vypsaná ANOVA na ověření významnosti dle F-hodnoty. ANOVA tvrdí, že model a tituly jsou statisticky nevýznamné, nicméně dle výstupu *S* by mělo být méně statisticky významné než *S/K* (a dokonce i méně významné než styl využití opce).

Vymazání odlehlých hodnot

Data před vymazáním

ticker	min	q1	median	q3	max
AAPL	-0.3736948	-0.0076819	0.0001582	0.0041181	0.3390972
AMD	-0.6154212	-0.0108625	0.0002307	0.0083084	0.2967886
AMZN	-0.2262697	0.0000482	0.0032476	0.0112189	0.2182972
GOOGL	-0.6033086	0.0004958	0.0073598	0.0213935	0.2831292
JNJ	-0.3712162	0.0003286	0.0033480	0.0113573	0.2850425
KO	-0.1759941	0.0001339	0.0021996	0.0081062	0.1745342
META	-0.4407167	-0.0031416	0.0004436	0.0056293	0.4305316
MSFT	-0.4926360	0.0002480	0.0052297	0.0168145	0.3386330
NFLX	-0.5742996	0.0003203	0.0041034	0.0151274	0.5724573
NVDA	-0.2339406	0.0000464	0.0023440	0.0145293	0.2200344
PG	-0.1799607	0.0001332	0.0023467	0.0092283	0.1678837
TSLA	-0.4674946	-0.0212378	-0.0009543	0.0021988	0.3323695
^RUT	-0.3326264	-0.0013374	0.0008961	0.0054200	0.3470238
^VIX	-0.5460372	-0.1351076	-0.0254084	0.0117656	0.3853825
^XDA	-0.0802413	-0.0301906	-0.0093713	0.0043543	0.0209063
^XDB	-0.0057359	0.0080747	0.0158229	0.0237110	0.0483671
^XDE	-0.0387247	-0.0016271	0.0028949	0.0103129	0.0327996
^XDN	-0.0369220	0.0022448	0.0072318	0.0177864	0.0623513



Proces vymazání

Data se odstraňují tak, aby chyba ocenění byla < 0.15 a aby strip plot ukazoval pouze přihléhlé hodnoty pro každý titul, za účelem vyřazení extrémů. Pro každý titul byla data omezena na hodnoty mezi 4% a 96% kvantilem, mimo tituly:

- Měnové indexy ($\hat{XD}_i$), které neomezujeme, jelikož jsou již korektní,
- NFLX, pro který je omezení 8% a 96%, z důvodu velkého počtu extrémních negativních hodnot.
- JNJ a PG, pro které je omezení 6% a 96%, ze stejného důvodu.

Titul $\hat{VIX}$ byl odstraněn kvůli zkreslení modelu kvůli vytváření samostatných clusterů s extrémy.

```
sort_mask <- function(dataset) {

  # Výchozí hodnoty - aplikováno na normální titul
  default_lower <- 0.04
  default_higher <- 0.97

  # Tituly, u kterých je nutno hodnoty omezit více
  quantiles <- list(
    NFLX = c(0.08, default_higher),
    JNJ = c(0.08, default_higher),
    PG = c(0.06, default_higher),
    MSFT = c(0.05, default_higher),
    KO = c(0.05, default_higher)
  )

  # Tituly, které nebudeme takto filtrovat
  banned <- c("^XDN", "^XDB", "^XDA", "^XDE")
  tickers <- c(unique(dataset$ticker))
  tickers <- setdiff(tickers, banned)

  # Základní maska
  # - Ze začátku předpokládáme že všechny opce jsou ok a chceme je ve výsledku
  mask <- rep(TRUE, nrow(dataset))

  for (cur_ticker in tickers) {
    # Získáváme pouze hodnoty pro daný ticker
    ticker_diff <- dataset |> filter(ticker == cur_ticker)

    # Výběr kvantilů pro omezení hodnot
    if (cur_ticker %in% names(quantiles)) {
      #cat(paste(cur_ticker, "found in quantiles\n"))
      quants <- quantiles[[cur_ticker]]
      low_q_const <- quants[1]
      high_q_const <- quants[2]
    } else {
      #cat(paste(cur_ticker, "not found in quantiles\n"))
      low_q_const <- default_lower
      high_q_const <- default_higher
    }

    # Zjištění hodnot v těchto kvantilech
    low_q <- quantile(ticker_diff$pricing_error, low_q_const)
    high_q <- quantile(ticker_diff$pricing_error, high_q_const)

    # Upravení masky
    # - Řádek se buď vztahuje k jinému titulu, nebo je v korektních hodnotách...
    # ...jinak je vynechán
    mask <- mask &
      ((dataset$pricing_error > low_q | dataset$ticker != cur_ticker) &
```

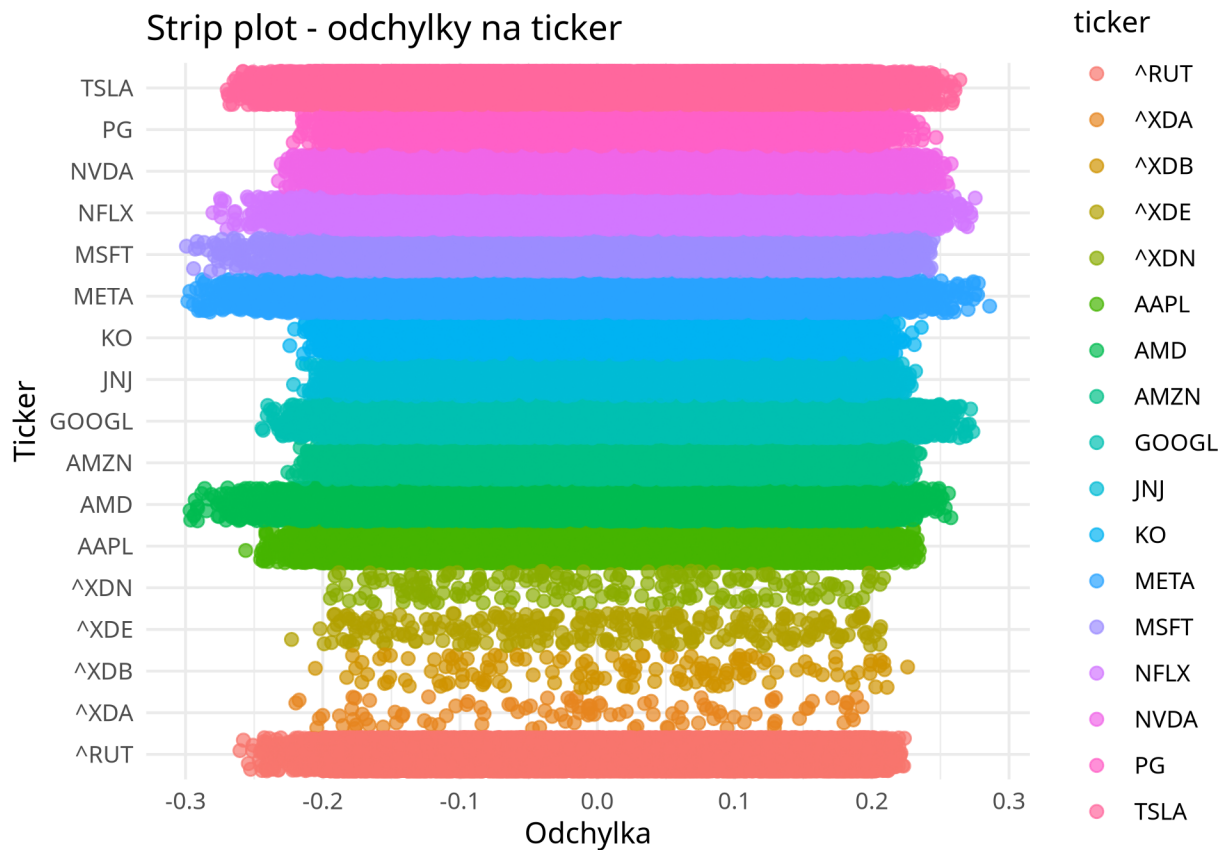
```

    (dataset$pricing_error < high_q | dataset$ticker != cur_ticker))
  }

  dataset[mask, ]
}

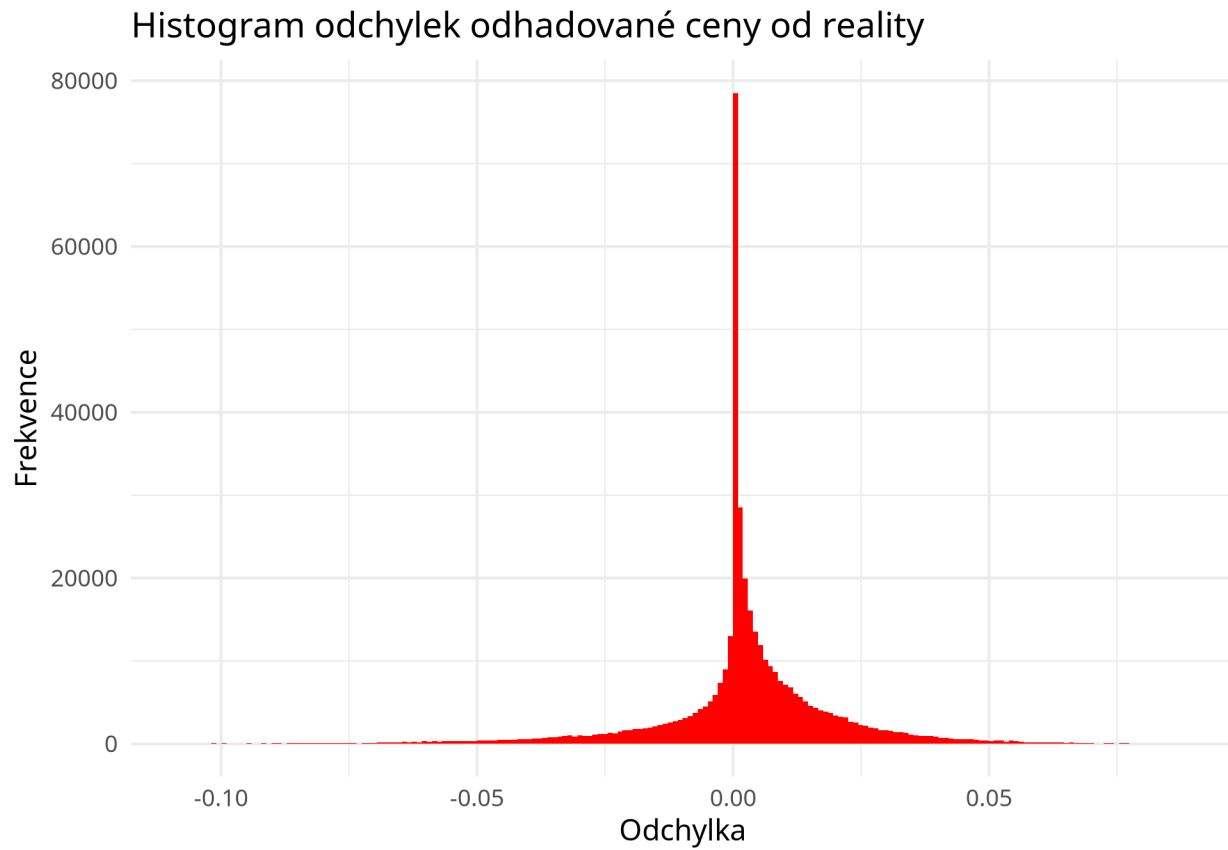
```

Data po vymazání



ticker	min	q1	median	q3	max
AAPL	-0.0566254	-0.0058287	0.0001619	0.0037798	0.0445029
AMD	-0.1073010	-0.0087902	0.0002527	0.0076089	0.0641908
AMZN	-0.0263148	0.0001126	0.0033769	0.0107601	0.0353304
GOOGL	-0.0482544	0.0007555	0.0076008	0.0204167	0.0787005
JNJ	-0.0225740	0.0006401	0.0038711	0.0112073	0.0361114
KO	-0.0322779	0.0002744	0.0023604	0.0075394	0.0450329
META	-0.1035577	-0.0021861	0.0004803	0.0051593	0.0876315
MSFT	-0.1025861	0.0004634	0.0055254	0.0162379	0.0472561
NFLX	-0.0832146	0.0007093	0.0047921	0.0149353	0.0817078
NVDA	-0.0338313	0.0000541	0.0024598	0.0133732	0.0587572
PG	-0.0266188	0.0003330	0.0025790	0.0088838	0.0510724
TSLA	-0.0709814	-0.0190257	-0.0007676	0.0018732	0.0680727
^RUT	-0.0631446	-0.0006373	0.0009368	0.0050719	0.0265446
^XDA	-0.0802413	-0.0301906	-0.0093713	0.0043543	0.0209063
^XDB	-0.0057359	0.0080747	0.0158229	0.0237110	0.0483671
^XDE	-0.0387247	-0.0016271	0.0028949	0.0103129	0.0327996

ticker	min	q1	median	q3	max
^XDN	-0.0369220	0.0022448	0.0072318	0.0177864	0.0623513



LMER model - základní

```
subset_args <- c(
  "S",
  "moneyness",
  "K",
  "r",
  "sigma",
  "q",
  "T",
  "am_eu",
  "call_put",
  "model",
  "ticker"
)
interactions <- c(
  "q:am_eu",
  "(1 | ticker)",
  "(1 | snapshot_month)"
)
rhs <- paste(c(subset_args, interactions), collapse = " + ")
```

```

formula <- as.formula(paste("pricing_error ~ ", rhs))

rmr_relative <- lmer(formula, data = train_data_rel)

## fixed-effect model matrix is rank deficient so dropping 2 columns / coefficients
summary(rmr_relative)

## Linear mixed model fit by REML ['lmerMod']
## Formula: pricing_error ~ S + moneyness + K + r + sigma + q + T + am_eu +
##      call_put + model + ticker + q:am_eu + (1 | ticker) + (1 |
##      snapshot_month)
##      Data: train_data_rel
##
## REML criterion at convergence: -1788573
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -6.9705 -0.3289  0.0572  0.4277  5.5928
##
## Random effects:
##      Groups          Name          Variance Std.Dev.
##      ticker          (Intercept) 4.792e-05 0.006923
##      snapshot_month (Intercept) 6.100e-07 0.000781
##      Residual                2.906e-04 0.017047
## Number of obs: 337180, groups:  ticker, 17; snapshot_month, 4
##
## Fixed effects:
##              Estimate Std. Error t value
## (Intercept) -2.875e-02  7.054e-03  -4.075
## S            1.131e-05  6.136e-07  18.441
## moneyness    -4.579e-05  3.624e-06 -12.634
## K            -7.610e-06  1.411e-07 -53.927
## r            1.726e+00  2.888e-02  59.769
## sigma       -7.960e-02  1.271e-03 -62.635
## q            -4.625e-01  5.101e-02  -9.065
## T            3.747e-03  4.202e-05  89.191
## am_eueuropean -2.782e-02  9.909e-03  -2.808
## call_putputs -3.102e-03  6.124e-05 -50.655
## modeluser_crr  2.225e-05  7.194e-05   0.309
## modeluser_mc   2.251e-05  7.190e-05   0.313
## ticker~XDA     -1.792e-02  1.008e-02  -1.777
## ticker~XDB      1.465e-02  1.003e-02   1.461
## ticker~XDE      1.279e-03  9.976e-03   0.128
## ticker~XDN      8.883e-03  1.001e-02   0.887
## tickerAAPL     -1.430e-02  9.799e-03  -1.459
## tickerAMD       9.425e-03  9.792e-03   0.963
## tickerAMZN     -4.984e-03  9.795e-03  -0.509
## tickerGOOGL     4.081e-04  9.799e-03   0.042
## tickerJNJ       3.819e-07  9.952e-03   0.000
## tickerKO       -4.653e-03  9.924e-03  -0.469
## tickerMETA     -7.751e-03  9.796e-03  -0.791
## tickerMSFT     -7.185e-03  9.807e-03  -0.733
## tickerNFLX     -2.254e-03  9.796e-03  -0.230

```

```
## tickerNVDA      2.990e-03  9.793e-03  0.305
## tickerPG       -3.612e-03  9.900e-03 -0.365

##
## Correlation matrix not shown by default, as p = 27 > 12.
## Use print(x, correlation=TRUE) or
##     vcov(x)         if you need it

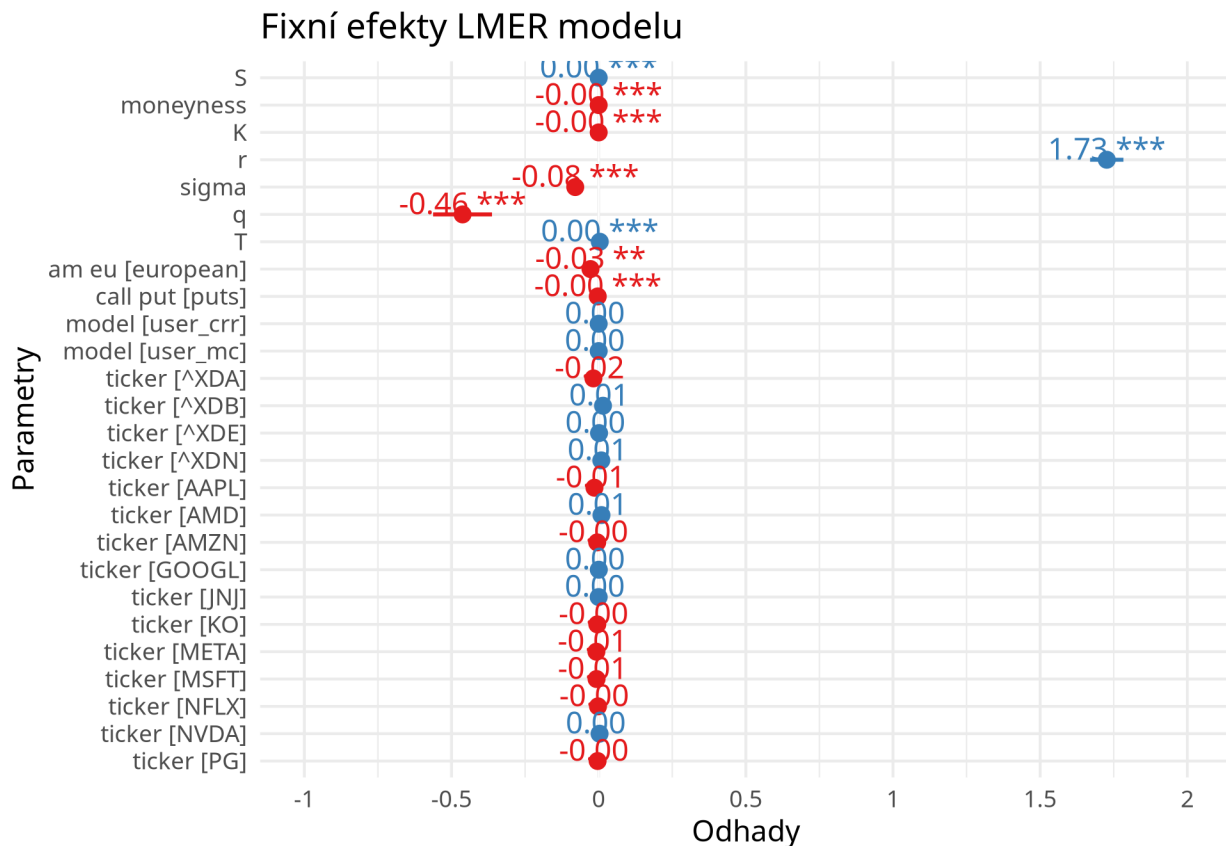
## fit warnings:
## fixed-effect model matrix is rank deficient so dropping 2 columns / coefficients
## Some predictor variables are on very different scales: consider rescaling.
## You may also use (g)lmerControl(autoscale = TRUE) to improve numerical stability.

r2(rmr_relative)

## Warning: Can't compute random effect variances. Some variance components equal
## zero. Your model may suffer from singularity (see `?lme4::isSingular`
## and `?performance::check_singularity`).
## Decrease the `tolerance` level to force the calculation of random effect
## variances, or impose priors on your random effects parameters (using
## packages like `brms` or `glmmTMB`).

## Random effect variances not available. Returned R2 does not account for random effects.

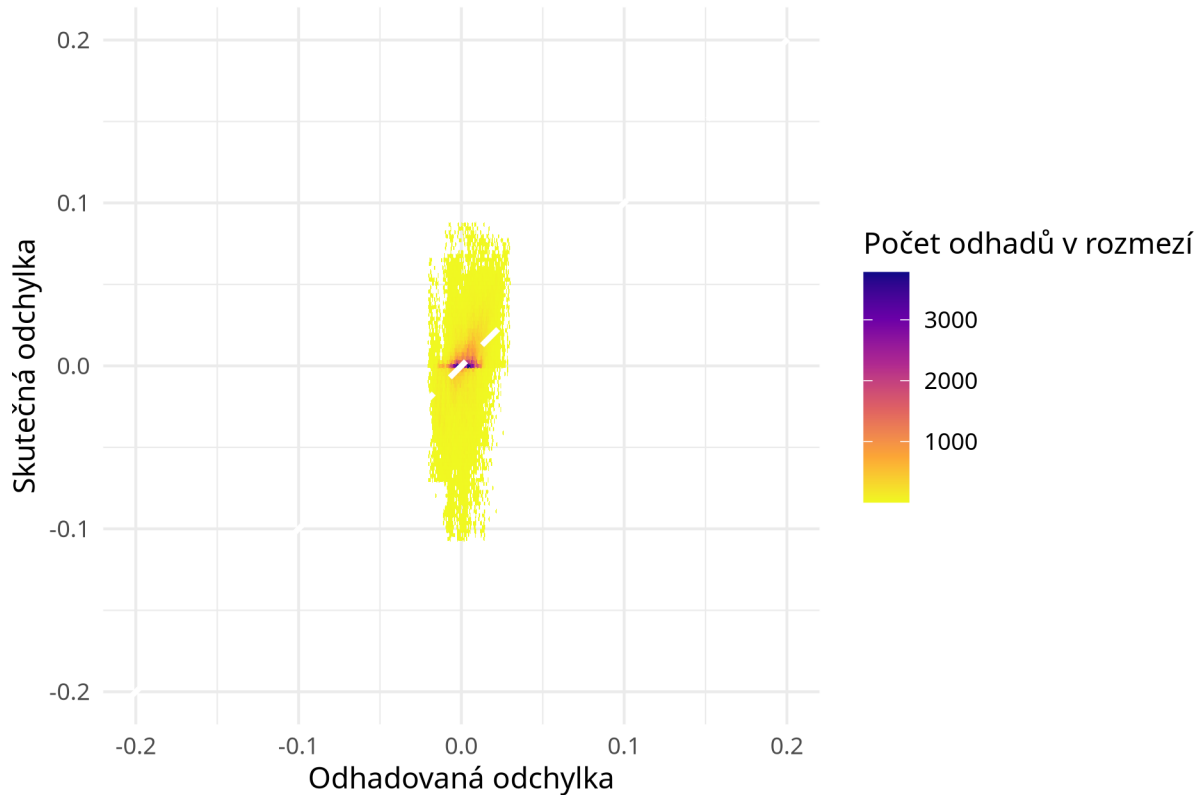
## # R2 for Mixed Models
##
## Conditional R2: NA
## Marginal R2: 0.164
```



Modely a tituly jsou statisticky nevýznamné, ostatní jsou statisticky silně významné. Některé tituly jsou

stasticiky slabě významné. Dle ANOVY potvrdíme zda je proměnná jako celek významná. Tituly jsou stále statisticky nevýznamné. Potenciální důvody mohou být parametry/skutečnosti mimo stahovaná data nebo vliv fixního efektu a náhodného efektu titulu na model.

Přesnost odhadu odchyly regresního modelu



```
anova(rmr_relative)
```

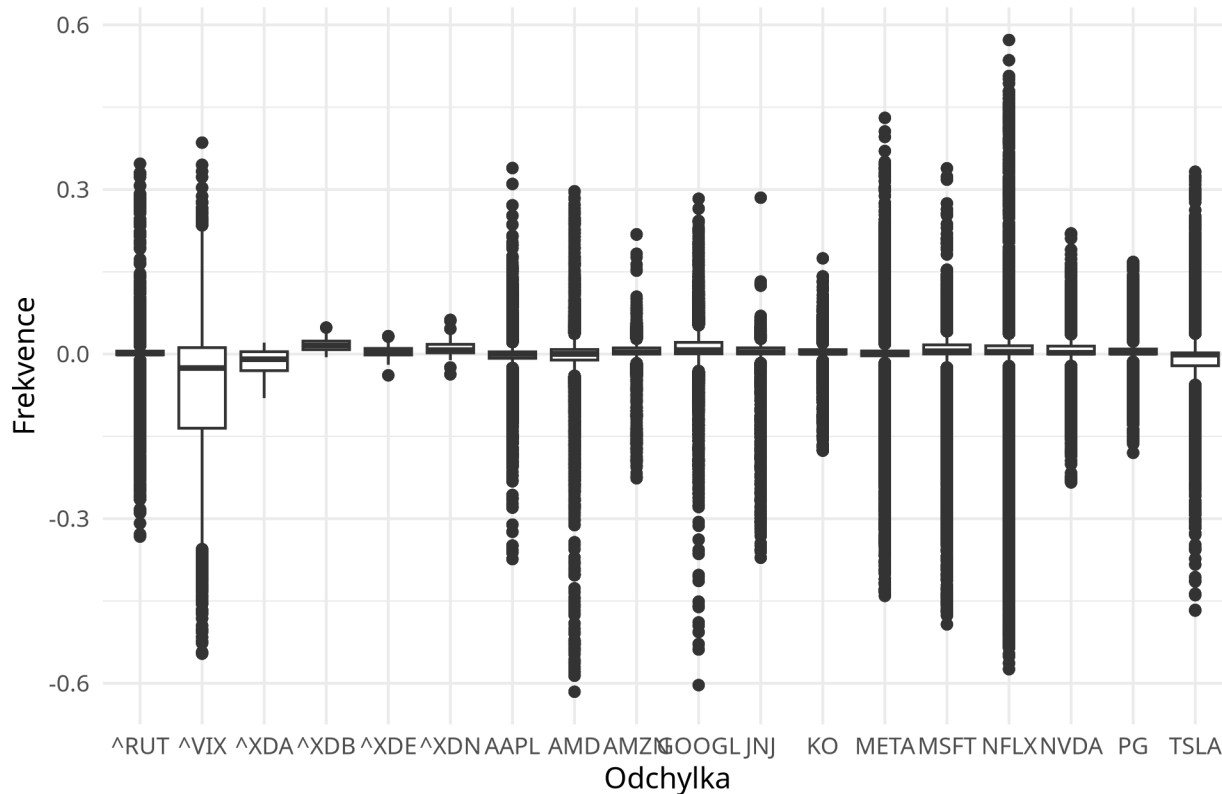
```
## Analysis of Variance Table
##          npar  Sum Sq Mean Sq  F value
## S              1 0.00057  0.00057    1.9626
## moneyness      1 0.00224  0.00224    7.7191
## K              1 0.50306  0.50306  1731.2109
## r              1 0.62103  0.62103  2137.1654
## sigma         1 1.08763  1.08763  3742.8993
## q             1 0.02453  0.02453   84.4279
## T             1 2.26014  2.26014  7777.8999
## am_eu         1 0.01170  0.01170   40.2648
## call_put      1 0.74524  0.74524  2564.6394
## model         2 0.00004  0.00002    0.0646
## ticker       15 0.00585  0.00039    1.3414
```

ANOVA opět potvrzuje nevýznamnost modelu a titulu, nicméně *S* je silně nevýznamné. U ostatních proměnných je potvrzena významnost.

Odstranění dle boxplotu

Experimentálně, zkusil jsem i odstranit outliery dle boxplotu (nechat pouze 1. - 3. kvartil).

Histogram odchylek odhadované ceny od reality



```
sort_mask <- function(dataset) {

  # Výchozí hodnoty - aplikováno na normální titul
  low_q_const <- 0.25
  high_q_const <- 0.75

  tickers <- c(unique(dataset$ticker))

  # Základní maska
  # - Ze začátku předpokládáme že všechny opce jsou ok a chceme je ve výsledku
  mask <- rep(TRUE, nrow(dataset))

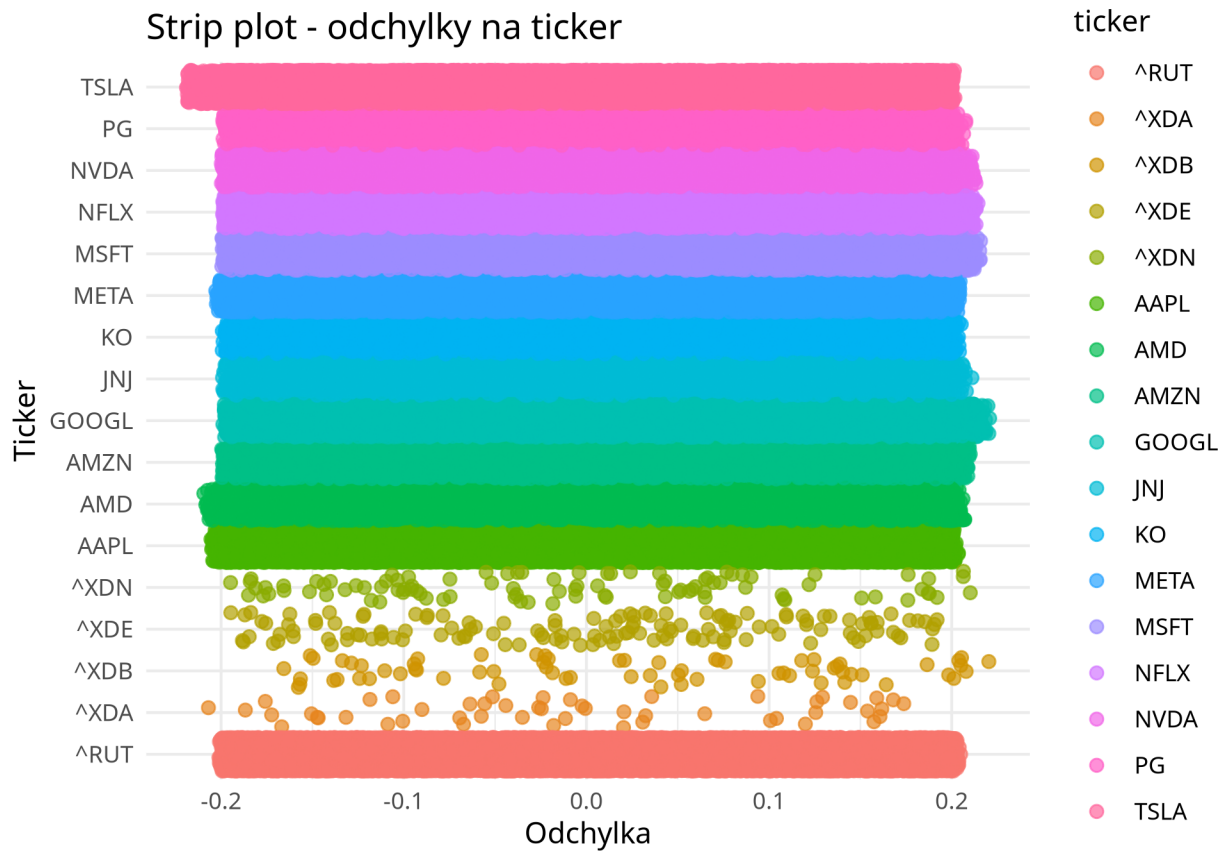
  for (cur_ticker in tickers) {
    # Ziskáváme pouze hodnoty pro daný ticker
    ticker_diff <- dataset |> filter(ticker == cur_ticker)

    # Zjištění hodnot v těchto kvantilech
    low_q <- quantile(ticker_diff$pricing_error, low_q_const)
    high_q <- quantile(ticker_diff$pricing_error, high_q_const)

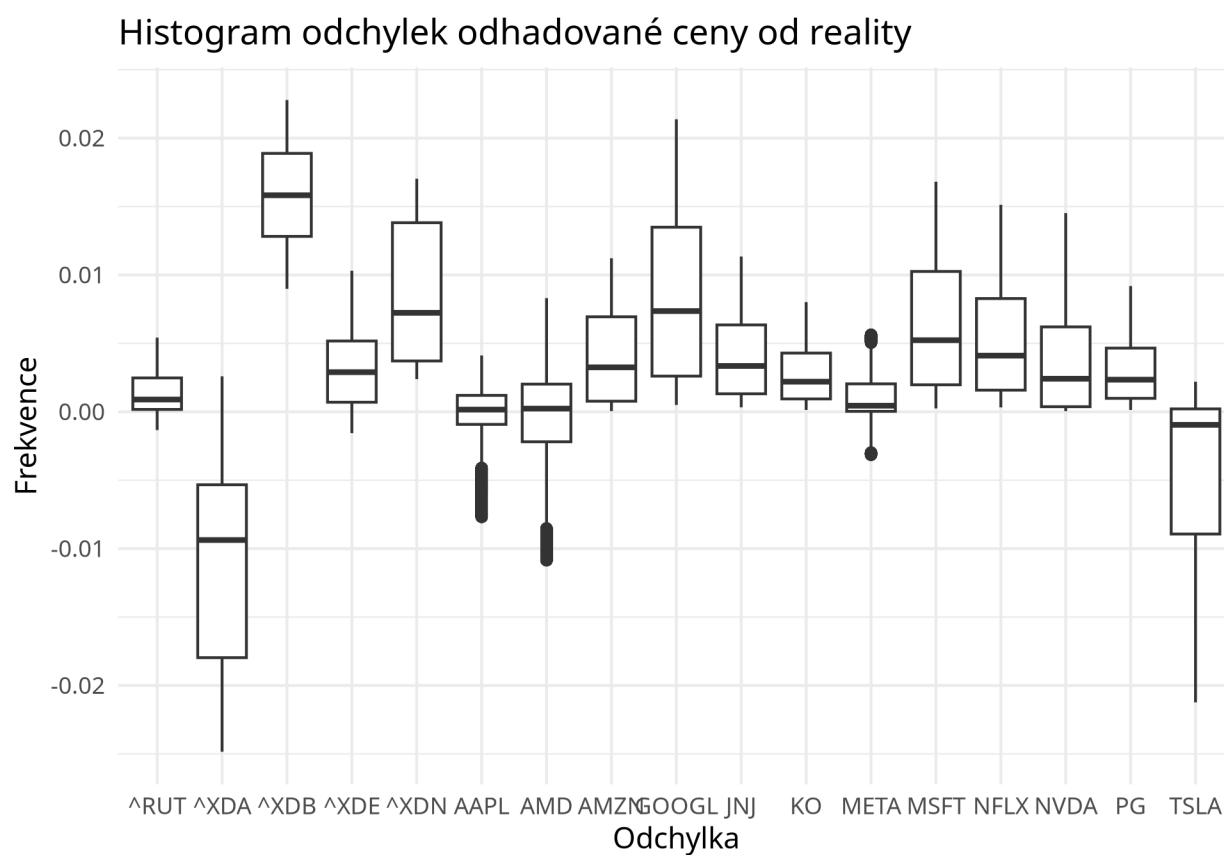
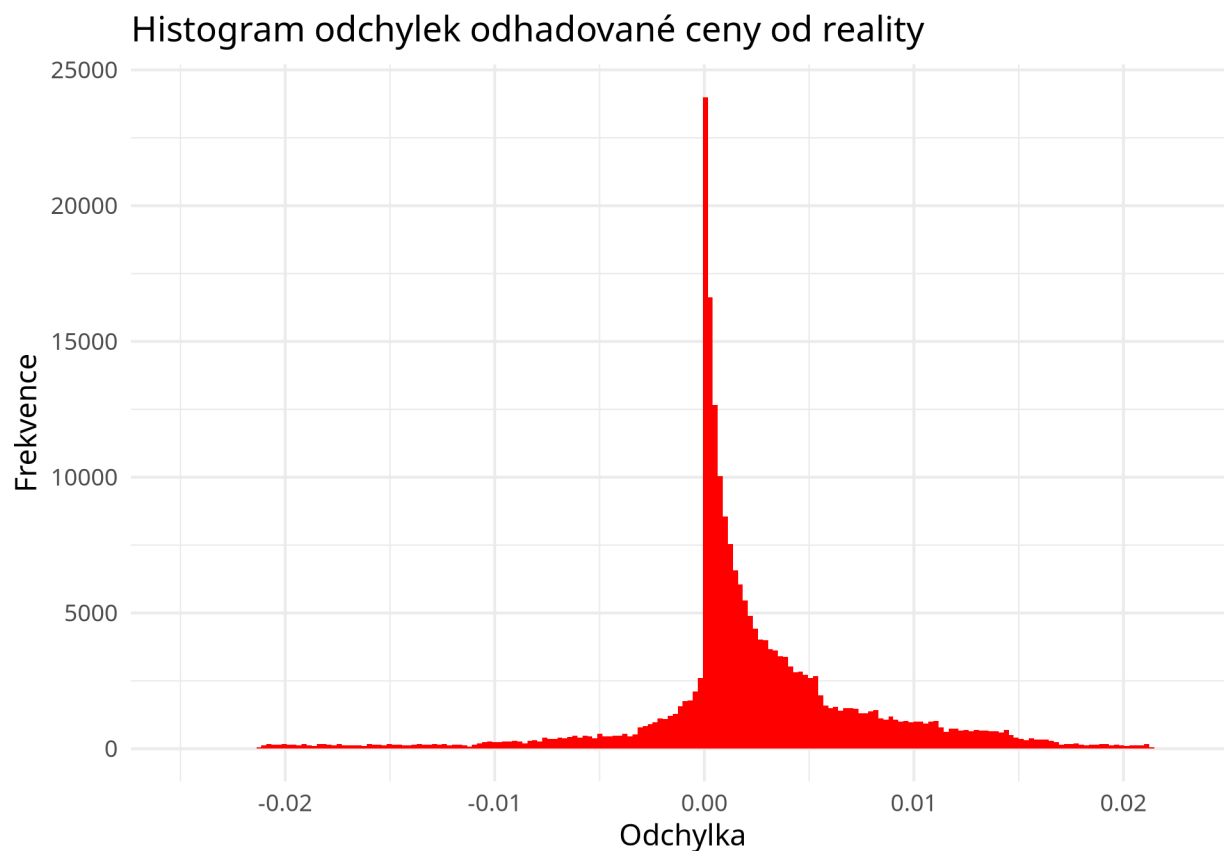
    # Upravení masky
    # - Řádek se buď vztahuje k jinému titulu, nebo je v korektních hodnotách...
    # ...jinak je vynechán
    mask <- mask &
      ((dataset$pricing_error > low_q | dataset$ticker != cur_ticker) &
       (dataset$pricing_error < high_q | dataset$ticker != cur_ticker))
  }
}
```

```
dataset[mask, ]
}
```

Data po vymazání



ticker	min	q1	median	q3	max
AAPL	-0.0076814	-0.0009202	0.0001582	0.0011995	0.0041172
AMD	-0.0108491	-0.0021917	0.0002307	0.0020213	0.0083058
AMZN	0.0000507	0.0007735	0.0032477	0.0069412	0.0112182
GOOGL	0.0004961	0.0026060	0.0073598	0.0134878	0.0213744
JNJ	0.0003286	0.0013143	0.0033480	0.0063479	0.0113459
KO	0.0001339	0.0009398	0.0021996	0.0042894	0.0080146
META	-0.0031406	0.0000328	0.0004436	0.0020414	0.0056242
MSFT	0.0002482	0.0019722	0.0052297	0.0102544	0.0168121
NFLX	0.0003205	0.0015810	0.0041034	0.0082731	0.0151268
NVDA	0.0000465	0.0003714	0.0024152	0.0062036	0.0145194
PG	0.0001332	0.0009830	0.0023467	0.0046530	0.0091910
TSLA	-0.0212363	-0.0089334	-0.0009543	0.0002113	0.0021987
^RUT	-0.0013370	0.0001705	0.0008961	0.0024738	0.0054195
^XDA	-0.0248434	-0.0179680	-0.0093713	-0.0053327	0.0025928
^XDB	0.0089830	0.0128165	0.0158229	0.0188826	0.0227831
^XDE	-0.0015601	0.0006982	0.0028949	0.0051723	0.0103078
^XDN	0.0023848	0.0037129	0.0072318	0.0138163	0.0170313



LMER model - základní

```
subset_args <- c(
  "S",
  "moneyness",
  "K",
  "r",
  "sigma",
  "q",
  "T",
  "am_eu",
  "call_put",
  "model",
  "ticker"
)
interactions <- c(
  "q:am_eu",
  "(1 | ticker)",
  "(1 | snapshot_month)"
)

rhs <- paste(c(subset_args, interactions), collapse = " + ")
formula <- as.formula(paste("pricing_error ~ ", rhs))

rmr_relative <- lmer(formula, data = train_data_rel)

## fixed-effect model matrix is rank deficient so dropping 2 columns / coefficients
summary(rmr_relative)

## Linear mixed model fit by REML ['lmerMod']
## Formula: pricing_error ~ S + moneyness + K + r + sigma + q + T + am_eu +
##      call_put + model + ticker + q:am_eu + (1 | ticker) + (1 |
##      snapshot_month)
##      Data: train_data_rel
##
## REML criterion at convergence: -1477524
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.6723 -0.5994 -0.0186  0.6123  3.2383
##
## Random effects:
##      Groups             Name             Variance Std.Dev.
##      ticker              (Intercept)  1.283e-05  0.0035812
##      snapshot_month      (Intercept)  1.557e-08  0.0001248
##      Residual                        1.766e-05  0.0042023
## Number of obs: 182325, groups:  ticker, 17; snapshot_month, 4
##
## Fixed effects:
##              Estimate Std. Error t value
## (Intercept)  5.726e-03  3.614e-03  1.585
## S            4.982e-06  2.079e-07  23.969
## moneyness    -1.273e-05  1.291e-06  -9.862
## K            -1.246e-06  4.901e-08 -25.429
```



```

## r          -1.118e-01  1.122e-02  -9.961
## sigma      -1.434e-02  4.295e-04 -33.392
## q          -8.509e-02  1.711e-02  -4.973
## T          9.172e-04  1.825e-05  50.258
## am_european -7.400e-03  5.091e-03  -1.454
## call_puts  1.342e-04  2.252e-05  5.962
## modeluser_crr -8.812e-06  2.410e-05  -0.366
## modeluser_mc -2.889e-05  2.410e-05  -1.199
## ticker^XDA  -4.469e-03  5.135e-03  -0.870
## ticker^XDB   2.171e-02  5.123e-03  4.238
## ticker^XDE   9.117e-03  5.108e-03  1.785
## ticker^XDN   1.592e-02  5.115e-03  3.112
## tickerAAPL   1.287e-03  5.067e-03  0.254
## tickerAMD    5.528e-03  5.065e-03  1.091
## tickerAMZN   5.857e-03  5.066e-03  1.156
## tickerGOOGL  9.703e-03  5.067e-03  1.915
## tickerJNJ    6.634e-03  5.100e-03  1.301
## tickerKO     5.549e-03  5.094e-03  1.089
## tickerMETA   2.387e-03  5.066e-03  0.471
## tickerMSFT   7.062e-03  5.068e-03  1.393
## tickerNFLX   7.595e-03  5.066e-03  1.499
## tickerNVDA   6.975e-03  5.065e-03  1.377
## tickerPG     5.569e-03  5.089e-03  1.094

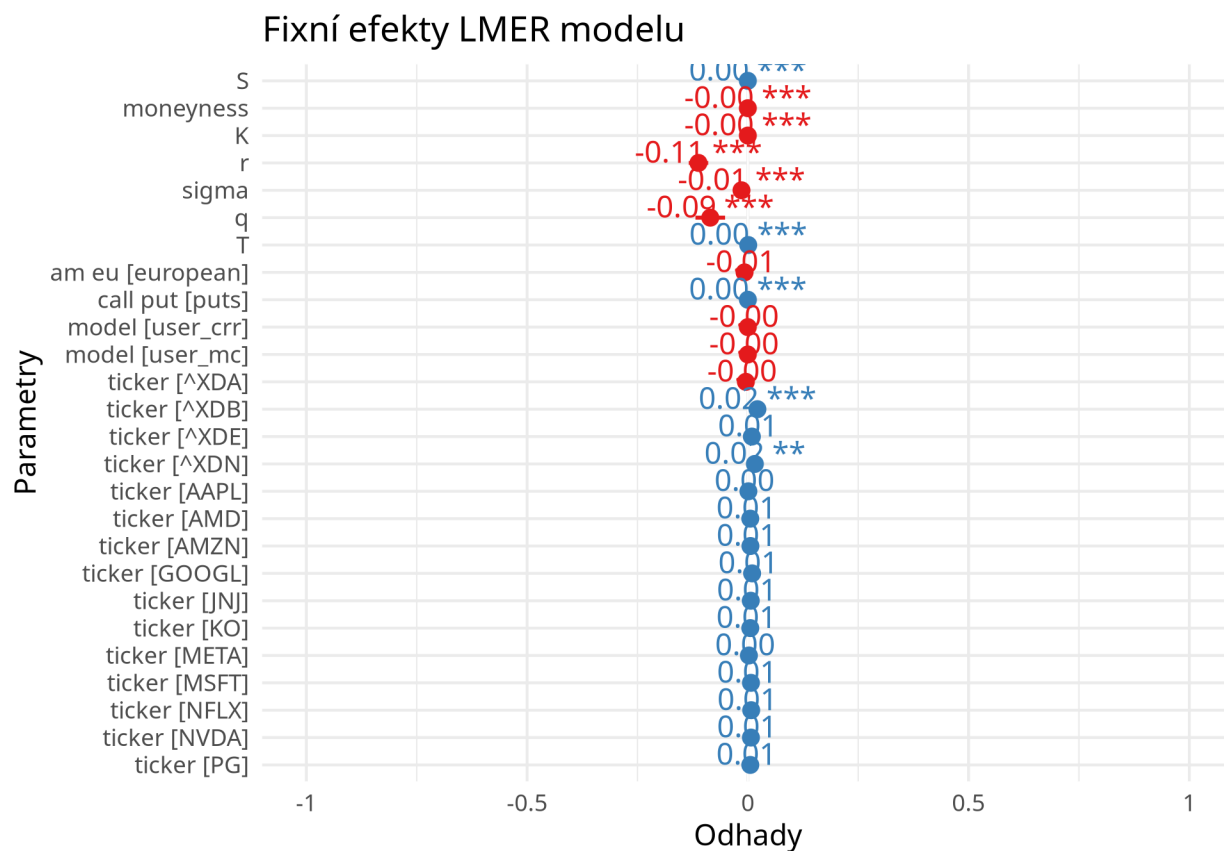
##
## Correlation matrix not shown by default, as p = 27 > 12.
## Use print(x, correlation=TRUE) or
##     vcov(x)           if you need it

## fit warnings:
## fixed-effect model matrix is rank deficient so dropping 2 columns / coefficients
## Some predictor variables are on very different scales: consider rescaling.
## You may also use (g)lmerControl(autoscale = TRUE) to improve numerical stability.
r2(rmr_relative)

## Warning: Can't compute random effect variances. Some variance components equal
## zero. Your model may suffer from singularity (see `?lme4::isSingular`
## and `?performance::check_singularity`).
## Decrease the `tolerance` level to force the calculation of random effect
## variances, or impose priors on your random effects parameters (using
## packages like `brms` or `glmmTMB`).

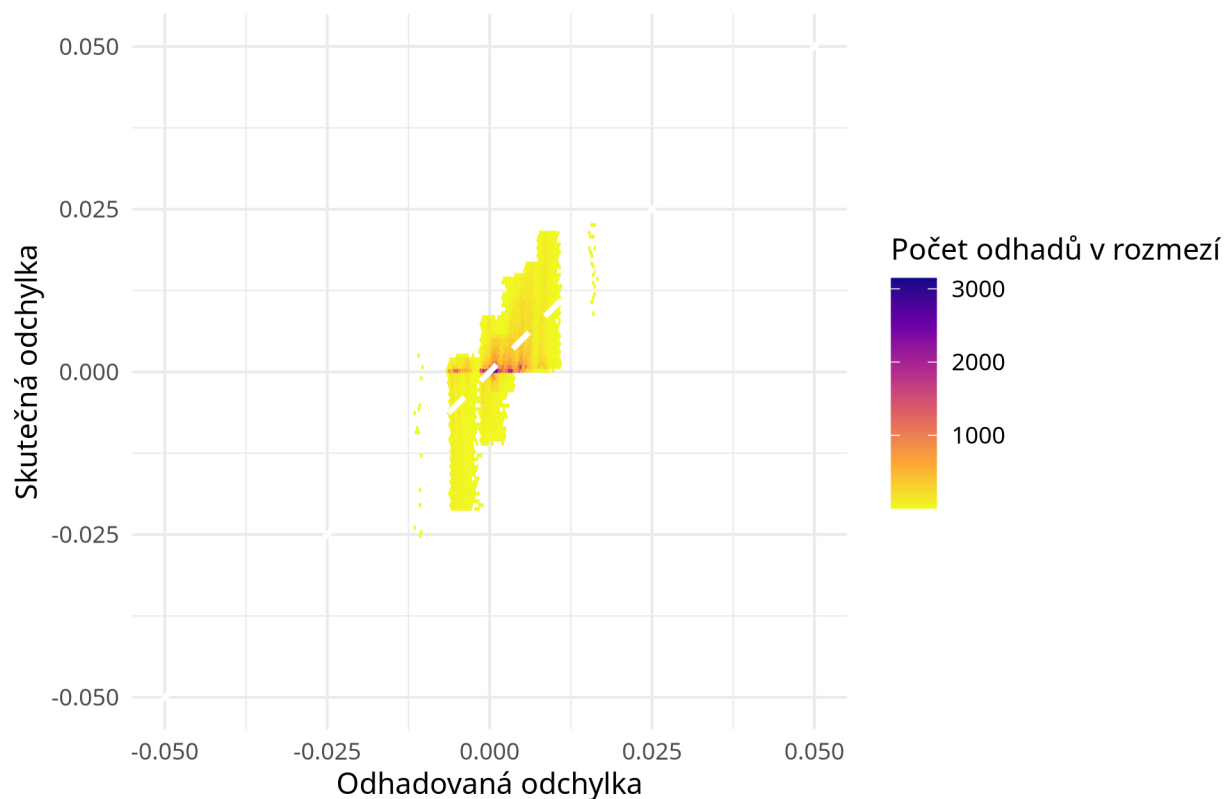
## Random effect variances not available. Returned R2 does not account for random effects.
## # R2 for Mixed Models
##
## Conditional R2: NA
## Marginal R2: 0.453

```



Modely, styly využití a tituly mimo ^XDB a ^XDN jsou statisticky nevýznamné. Zmíněné měnové fondy a ostatní parametry jsou statisticky významné. Nevýznamnost titulů potvrdíme v ANOVA analýze.

Přesnost odhadu odchyly regresního modelu



```
anova(rmr_relative)
```

```
## Analysis of Variance Table
##           npar   Sum Sq Mean Sq  F value
## S              1 0.005308 0.005308   300.6048
## moneyness      1 0.000047 0.000047     2.6410
## K              1 0.020088 0.020088  1137.5378
## r              1 0.011970 0.011970   677.8026
## sigma         1 0.018012 0.018012 1019.9656
## q              1 0.000301 0.000301    17.0208
## T              1 0.045501 0.045501 2576.5843
## am_eu         1 0.000086 0.000086     4.8719
## call_put      1 0.000630 0.000630    35.6547
## model         2 0.000027 0.000013     0.7530
## ticker       15 0.000748 0.000050     2.8246
```

Model a tituly jsou nevýznamné. Styl využití je na hranici významnosti. S/K je také na hranici významnosti, S je oproti ostatním modelům silně významnější.